

Fady Nabil

Computer Science Bachelor's Graduate /

+20-1226773371 · fadynf05@gmail.com · [LinkedIn](#) · [Github](#) · [Credly](#) · [Personal Website](#)

Hadayek El Kobba, Cairo, Egypt

ABOUT ME

Senior Computer Science student (AI Major) at Misr International University, graduating June 2026. I build end-to-end AI systems — from ML pipelines and cloud infrastructure on GCP to Flutter mobile apps and backend APIs — and have built systems at ElSewedy Electric and Al Ahli Bank of Kuwait. Currently leading the AI and system integration track on ReVerie, a VR-based Alzheimer's therapy platform, as my graduation project. Looking for a full-time AI/ML or backend engineering role where I can ship real systems at scale.

TECHNICAL SKILL SET

- **Languages** — Python · C++ · C# · Java · PHP · JavaScript · SQL · Flutter
 - **AI / ML** — PyTorch · TensorFlow · HuggingFace Transformers · NLP Pipelines · Computer Vision · 3D Reconstruction · YOLO · OpenCV · RAG · LLMs · Prompt Engineering
 - **Backend & Databases** — FastAPI · Node.js · Express.js · PostgreSQL · MongoDB · Supabase · pgvector · REST API Design
 - **Cloud & DevOps** — GCP · Azure · Linux · Git / GitHub
 - **Web** — HTML · CSS · JavaScript · PHP
-

EXPERIENCE

Artificial Intelligence Intern

ElSewedy Electric Headquarters

July 2025 - August 2025

- Built a production HR chatbot using a hand-rolled **RAG** pipeline (**PostgreSQL** + **pgvector**, **OpenAI API**, **Streamlit**), achieving 100% policy query coverage across the company's full HR documentation.
- Engineered a semantic PDF chunking system using the **unstructured** library and **OCR**, cutting retrieval latency by structuring document splits around meaning rather than fixed token windows.

Applications Developer Intern

Al Ahli Bank of Kuwait - Egypt Headquarters

August 2024 - September 2024

- Built a loan and lease payment scheduling system in **C#** and **SQL Server**, automating end-to-end payment tracking for a banking operation processing hundreds of transactions monthly.
 - Led requirements gathering through stakeholder interviews, translated business rules into system logic, and delivered the full implementation solo within a 1-month internship window.
 - Automated payment tracking and scheduling workflows, reducing manual processing errors by 80% and eliminating an estimated 15+ hours of weekly manual work.
-

CERTIFICATIONS

- IBM SkillsBuild — AI Fundamentals with Capstone (Feb 2025)
- IBM SkillsBuild — Generative AI for Software Development (Feb 2025)
- IBM SkillsBuild — Generative AI for Data Insights (Feb 2025)

PROJECTS

Reverie – AI-Powered Therapeutic Platform for Alzheimer's Patients

September 2025 - Current

Graduation Project · Flutter · GCP · Supabase · 3D Reconstruction · FastAPI · Linux Server

- Led **system architecture and integration** of a VR-based Alzheimer's therapy platform combining cognitive stimulation therapy with **AI-generated 3D environments**, built across a 5-person team.
- Integrated the full AI pipeline: a **7-stage image-to-3D reconstruction system** (depth estimation, segmentation, object + human reconstruction, etc...) averaging ~10-15 min per scene on GCP L4 GPU.
- Migrated all AI services and backend infrastructure to **GCP**, reducing local setup overhead by ~70% and enabling parallel AI inference across the team.
- Built the **Flutter** mobile app end-to-end (**Riverpod, Supabase, Agora video, biometric auth**) and integrated it with the AI pipeline, backend APIs, and Unreal Engine VR system.

Extractive Text Summarization App

May 2025 - June 2025

NLP · HuggingFace · BERT · BART · T5 · BiLSTM · XGBoost

- Built and benchmarked **9 summarization models** (KNN, Logistic Regression, XGBoost, BERTSum, BiLSTM, LSTM, MiniLM, BART, T5) across BBC and IMDB datasets.
- Designed a full preprocessing pipeline — text cleaning, tokenization, and attention-based sequence modeling — feeding into all 9 model variants.
- Achieved 100% accuracy and ROUGE-L 0.886 on BBC (KNN), and 92% accuracy with ROUGE-L 0.709 on IMDB (Logistic Regression) — outperforming several transformer baselines on extractive tasks.

Football Analysis System

Jan 2025

Deep Learning · YOLOv8 · ByteTrack · KMeans · Python

- Built a real-time football analysis system using **YOLOv8 + ByteTrack** for multi-player detection and tracking across full match videos.
- Applied **KMeans clustering** on jersey color features to automatically assign players to teams — no manual labeling required.
- Generated annotated output videos with per-frame player tracks, team labels, and ball trajectory overlays.

Car Recommendation and Comparison Web Application

Oct 2024 - Dec 2024

Software Engineering · PHP · MySQL · HTML/CSS/JS · OpenAI API

- Collaborated on Apex, a full-stack car recommendation platform, building the backend in PHP with MySQL and integrating OpenAI's ChatGPT API for AI-powered personalized recommendations.
- Developed a persona-based quiz engine to match users to vehicles based on preferences, and a social system supporting reviews and likes.
- Delivered full-stack features across backend, frontend, and database layers within a 2-month team sprint.

Meal Planner and Recommendation Web App

Feb 2024 - July 2024

Backend · MongoDB · Express.js · Node.js · JavaScript

- Built a full-stack subscription-based meal platform using the MEN stack, handling user authentication, subscription state management, and personalized meal recommendation logic on the backend.
- Designed and implemented a RESTful API layer in Express.js serving dynamic meal recommendations based on user dietary preferences and order history.
- Structured MongoDB collections for users, meals, subscriptions, and orders — optimizing query patterns for a preference-driven recommendation engine.

EDUCATION

Bachelor in Computer Science - AI Minor

Misr International University

2022 - 2026 (Expected)

GPA: 3.16 / 4.0

American High School Diploma

St.John American International School

2008 - 2022

GPA: 3.5 / 4.0

EXTRACURRICULAR ACTIVITIES

Teaching Assistant in MIU

Co-assisted in 4 core CS courses: Intro to CS, Computer Programming, Data Structures & Algorithms, and OOP — supporting students in labs, grading, and concept reinforcement.